4) $\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$		Rules of Conjunction
5) $\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$[(p \vee q) \wedge \neg p] \to q$	Rule of Disjunctive Syllogism
6) $\begin{array}{l} \neg p \to F \\ \hline \therefore p \end{array}$	$(\neg p \to F_0) \to p$	Rule of Contradiction
7) $\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \to p$	Rule of Conjunctive simplification
8) $\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \to p \vee q$	Rule of Disjunctive Amplification
9) $\begin{array}{l} p \wedge q \\ p \to (q \to r) \\ \hline \therefore r \end{array}$	$[(p \wedge q) \wedge [p \to (q \to r)]] \to r$	Rule of Conditional Proof
10) $\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore (p \vee q) \to r \end{array}$	$[(p \to r) \wedge (q \to r)] \to [(p \vee q) \to r]$	Rule for Proof by Cases
11) $\begin{array}{l} p \to q \\ r \to s \\ p \vee r \\ \hline \therefore q \vee s \end{array}$	$[(p \to q) \wedge (r \to s) \wedge (p \vee r)] \to (q \vee s)$	Rule of the Constructive Dilemma
12) $\begin{array}{l} p \to q \\ r \to s \\ \neg q \vee \neg s \\ \hline \therefore \neg p \vee \neg r \end{array}$	$[(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)] \to (\neg p \vee \neg r)$	Rule of the Destructive Dilemma

$\exists x\, p(x)$	For some (at least one) a in the universe, $p(a)$ is true.	For every a in the universe, $p(a)$ is false.
$\forall x\, p(x)$	For every replacement a from The universe, $p(a)$ is true.	There is at least one replacement a from the universe for which $p(a)$ is false.
$\exists x \neg p(x)$	For at least one choice a in The universe, $p(a)$ is false, so Its negation $\neg p(a)$ is true.	For every replacement a in the universe, $p(a)$ is true.

$\forall x \neg p(x)$	For every replacement *a* from The universe, *p(a)* is false and its negation $\neg p(a)$ is true	There is at least one replacement *a* from the universe for which $\neg p(a)$ is false and $p(a)$ is true.

$\neg [\forall x\, p(x)] \Leftrightarrow \exists x \neg p(x)$
$\neg [\exists x\, p(x)] \Leftrightarrow \forall x \neg p(x)$
$\neg [\forall x \neg p(x)] \Leftrightarrow \exists x \neg\neg p(x) \Leftrightarrow \exists x\, p(x)$
$\neg [\exists x \neg p(x)] \Leftrightarrow \forall x \neg\neg p(x) \Leftrightarrow \forall x\, p(x)$

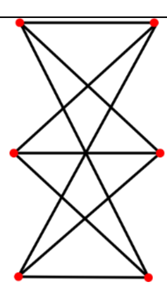
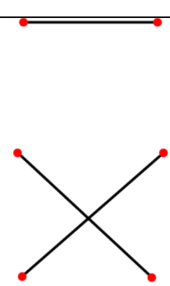
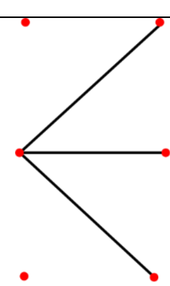
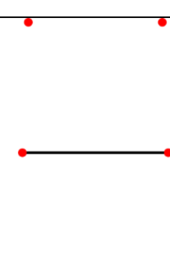
$\exists x\, [P(x) \vee Q(x)] \equiv \exists x\, P(x) \vee \exists x\, Q(x)$
$\forall x\, [P(x) \vee Q(x)] \equiv \forall x\, P(x) \vee \forall x\, Q(x)$
$\exists x\, [P(x) \wedge Q(x)] \to \exists x\, P(x) \wedge \exists x\, Q(x)$
$\forall x\, [P(x) \wedge \forall\, Q(x)] \to \forall x\, [P(x) \vee Q(x)]$
$\forall x\, [P(x) \to Q(x)] \to \forall x\, P(x) \to \forall x\, Q(x)$
$\forall x\, [P(x) \leftrightarrow Q(x)] \to \forall\, P(x) \leftrightarrow \forall x\, Q(x)$
$(\forall x)\, P(x) \wedge (\forall x)\, Q(x) \Leftrightarrow (\forall x)\, [P(x) \wedge Q(x)]$
$(\forall x)\, P(x) \wedge (\forall x)\, Q(x) \Leftrightarrow (\forall x)\, (\forall y)\, [P(x) \vee Q(y)]$
$(\exists x)\, P(x) \vee (\exists x)\, Q(x) \Leftrightarrow (\exists x)\, (\exists y)\, [P(x) \wedge Q(y)]$
$(\exists x)\, P(x) \vee (\exists x)\, Q(x) \Leftrightarrow (\exists x)\, [P(x) \vee Q(y)]$
$(\forall x)\, P(x) \wedge (\exists x)\, Q(x) \Leftrightarrow (\forall x)\, (\exists y)\, [\, P(x) \wedge Q(y)]$
$(\forall x)\, P(x) \vee (\exists x)\, Q(x) \Leftrightarrow (\forall x)\, (\exists y)\, [\, P(x) \vee Q(y)]$
$A \vee (\forall x)\, P(x) \Leftrightarrow (\forall x)\, [A \vee P(x)]$
$A \vee (\exists x)\, P(x) \Leftrightarrow (\exists x)\, [A \vee P(x)]$
$A \wedge (\forall x)\, P(x) \Leftrightarrow (\forall x)\, [A \wedge P(x)]$
$A \wedge (\exists x)\, P(x) \Leftrightarrow (\exists x)\, [A \wedge P(x)]$

Statement	When True?	When False?
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true	There is an x such that $P(x, y)$ is false for every y.
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y.	For every x there is a y for which $P(x, y)$ is false.

$\exists x\exists y P(x, y)$ $\exists y\exists x P(x, y)$	The is a pair x, y for which $P(x, y)$ is true	$P(x, y)$ is false for every pair x, y

$$
\begin{tikzcd}
\forall x\forall y \arrow[r, leftrightarrow] \arrow[d] & \forall y\forall x \arrow[d] \\
\exists y\forall x \arrow[d] & \exists x\forall y \arrow[d] \\
\forall x\exists y \arrow[d] & \forall y\exists x \arrow[d] \\
\exists y\exists x \arrow[r, leftrightarrow] & \exists x\exists y
\end{tikzcd}
$$

Rule of Inference	***Name***
$\dfrac{\forall x P(x)}{\therefore P(c)}$	Universal instantiation
$\dfrac{P(c) \text{ for an arbitary } c}{\therefore \forall x P(x)}$	Universal generalization
$\dfrac{\exists x P(x)}{\therefore \text{ for some element } c}$	Existential instantiation
$\dfrac{P(c) \text{ for some elemet } c}{\therefore \exists x P(x)}$	Existential generalization

<table>
<tr><td colspan="4">Number of non-equivalent propositional function possible with ‘n’ propositional variable</td></tr>
<tr><td colspan="4">2^{2^n}</td></tr>
<tr><td colspan="4">Nested Quantifier</td></tr>
<tr><td>$\forall x\forall y$</td><td>$\forall x\exists y$</td><td>$\exists y\forall x$</td><td>$\exists x\exists y$</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="2">English statements</td><td colspan="2">Logical Expressions</td></tr>
</table>

All graphs are connected	$\forall x[G(x) \to C(x)]$
Not all graphs are connected	$\neg\forall x[G(x) \to C(x)]$
All graphs are not connected ≡ No graphs are connected	$\forall x[G(x) \to \neg C(x)] \equiv \forall x[G(x) \to \neg C(x)]$

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